Using BigQuery ML to Train a Linear Regression Model

Use SQL and BigQuery ML for power-production regression, learning how correlation, preprocessing, explainability, and neural-network nonlinearity affect the pipeline.

Original study notes based on Gwendolyn Stripling and Michael Abel, First edition, 2023. Book pp. 147–192. Source APIs, services, experiments, and model comparisons reflect the book's publication period. These are explanatory summaries, not verbatim reproductions or a claim that every source detail is retained.

Load + clean in SQL → Explore relationships → Train BigQuery ML → Explain + validate → Compare linear and neural models

Detailed Notes

Power plant objective and SQL data preparation

Source: book pp. 147–159 / PDF p. 163

Predict net hourly power output from weather/plant variables. Understand the row meaning, units, ranges, and prediction timing. Load CSV into cloud storage/BigQuery, verify schema and row counts, and clean malformed or missing values with explicit SQL.

Use SQL to profile distributions and identify outliers, then validate transformed data. Keep credentials and project identifiers out of shared notebooks. Track dataset versions and ensure train/evaluation rows are separated before model fitting.

Linear regression, features, correlation

Source: book pp. 160–174 / PDF p. 176

Regression predicts a continuous energy-output label. Inspect feature-label relationships and predictor-predictor correlations. Multicollinearity can destabilize coefficient estimates and make individual effects difficult to interpret even when overall predictions remain usable.

$$\hat{y} = \beta_0 + \Sigma\, \beta_j x_j$$

Correlation is not causal influence. Use scatterplots and heat maps to generate hypotheses, not as the only feature-selection rule. Avoid leakage, keep transformations consistent, and compare errors on held-out data.

CREATE MODEL and explainability

Source: book pp. 175–182 / PDF p. 191

BigQuery ML expresses model training through SQL such as CREATE MODEL, specifying model type, label column, options, and a SELECT query. Separate training data from evaluation; retain model metadata and query versions for reproducibility.

Explainability tools can estimate how features contribute to predictions. Global importance and local attribution answer different questions and can be affected by correlated predictors. Explanations describe model behavior under a method; they do not prove physical causation. Use exercises to test edge cases and interpret the scale of predictions.

Neural networks and nonlinearity

Source: book pp. 183–192 / PDF p. 199

A neural network stacks learned affine transformations and activation functions. Nonlinear activations allow the network to model relationships that a single linear equation cannot. Without nonlinearities, stacked dense layers collapse into one linear transform.

> $h = \phi(Wx+b)$

Train a BigQuery ML DNN regressor using suitable features, loss, and evaluation. Compare with linear regression on the same holdout and metric. More flexibility can overfit and reduce interpretability; it does not imply a better result. The chapter's Cloud Shell section illustrates inspecting cloud files and command output, rather than a new modeling method.

Revision Check

1. What does multicollinearity do to coefficient interpretation?
2. Why does a DNN need nonlinear activations?
3. What does explainability reveal and not prove?
4. How should the linear and neural models be compared?

SQL can simplify end-to-end modeling near warehouse data, but it does not remove the need for clean splits, diagnostics, and careful interpretation.

Source links refer to the supplied PDF. Every contents-page topic is mapped in coverage.md. Consult the book for complete implementations, figures, exercises, citations, and experiment evidence.